

Deepak Mallampati

APPLIED AI ENGINEER - AGENTIC AI & RETRIEVAL SYSTEMS

(330) 554-0770 | deepakmallampati00@gmail.com | linkedin.com/in/deepak-mallampati |
US-based, F-1 OPT - authorized to work in the US now

PROFESSIONAL SUMMARY

Applied AI engineer with 4+ years building production **agentic AI, retrieval systems**, and backend data pipelines on distributed infrastructure. Ships reliable LLM agents and RAG systems - **42% token reduction** via conductor-subagent orchestration, **40% hallucination reduction** and **92% precision** on large-scale retrieval, and **12 microservices** at **100,000+ requests/day** (p95 800ms). Strong software-engineering fundamentals across Python, Kafka, Docker/Kubernetes, and AWS.

SKILLS

Agentic AI & AI Agents • Retrieval Systems (RAG) • ML Models • Data Pipelines • Backend Services & APIs • Distributed Systems • Python • Kafka • Cloud (AWS)

Languages & Backend: Python, Java, JavaScript/TypeScript, SQL | FastAPI, REST APIs, microservices

Agentic AI & LLMs: Multi-Agent Systems | LangChain/LangGraph | AI Agents | Prompt Engineering | GPT-4o, Claude, Gemini, LLaMA

Retrieval & ML: RAG Architecture | FAISS, Pinecone, Weaviate | Cross-Encoder Re-ranking | PyTorch, TensorFlow | NER, Classification

Infrastructure & Data: Kafka | Docker, Kubernetes | AWS (EC2, S3, Lambda, Bedrock, SageMaker) | PostgreSQL, MongoDB, Redis | CI/CD

MLOps & Reliability: MLflow, W&B | Model Monitoring & Observability | RAGAS, BERTScore | Terraform

PROFESSIONAL EXPERIENCE

Progress Solutions

Jul 2025 - Present

AI Engineer

USA

- Architected production-grade **agentic LLM systems** on a conductor-subagent orchestration framework, decomposing complex objectives into context-scoped subagents that execute autonomously - reducing token consumption by **42%** while sustaining task accuracy across multi-step workflows.
- Engineered high-performance **retrieval-augmented generation (RAG)** pipelines over large-scale healthcare document corpora, combining dense vector retrieval with cross-encoder re-ranking to lift answer relevance and materially reduce hallucination on domain-specific queries.
- Established a structured **LLM evaluation and monitoring framework** (RAGAS, BERTScore, custom domain metrics) paired with latency and accuracy dashboards to benchmark model iterations pre-deployment and surface regressions before release.
- Built fault-tolerant **automation infrastructure** featuring scheduled batch orchestration and exponential-backoff retry logic, improving pipeline reliability and reducing failed production runs by **60%**.
- Optimized inference cost and latency through prompt compression, semantic caching, and model routing, sustaining SLA targets while reducing per-query overhead.

Vanguard

Jan 2024 - Jan 2025

AI Engineer Intern

USA

- Developed and enhanced **AI agents and backend services** for Vanguard's internal AI platform, integrating with production data pipelines and observability tooling across **25+ AI agents** and workflows.
- Built **12 APIs and microservices** powering new AI capabilities, reducing p95 latency from 1.5s to **800ms** and accelerating feature integration by **40%**.
- Optimized LLM prompts and evaluated **GPT-4o, Claude Sonnet, and Gemini** models, improving task success rates by **27%** on internal evaluation datasets.
- Engineered safeguards and content controls for secure, policy-compliant AI responses, reducing unsafe outputs by **42%** while maintaining response quality.
- Partnered with product, platform, and data engineering teams to deliver scalable LLM applications supporting **100,000+ requests daily** across 20+ internal teams.

Kent State University

Oct 2023 - Jan 2024

ML & Gen AI Research Assistant

USA

- Built a **privacy-preserving ML pipeline** on a clinical hospital-readmission dataset (k-anonymity, l-diversity, Laplace differential privacy), reducing re-identification risk from **3.38% to 0.32%** while retaining **99% of baseline predictive performance**.
- Fine-tuned **Qwen3-27B via 4-bit QLoRA** on an NVIDIA H100 using a curated instruction dataset for controllable long-form generation; authored a peer-review-ready IEEE-format paper.

Emerson

Jun 2022 - Apr 2023

ML Trainee

India

- Developed and trained supervised **regression and classification models** for equipment-failure prediction, anomaly detection, and capacity forecasting on operational data.
- Fine-tuned **BERT and RoBERTa** models for entity extraction and text classification, achieving **89% F1-score** on custom NER tasks.

PROJECTS

MangaLens

A shipped Chrome extension (live on the Chrome Web Store) delivering real-time AI translation of manga and webtoons across **29+ sites**, turning a multi-step manual translation chore into a single inline action.

career-ops

An autonomous, AI-driven job-search pipeline that scans listings, scores fit, and generates tailored applications overnight without supervision - compressing hours of repetitive work into an automated workflow.

Privacy-Preserving Clinical ML Pipeline

A framework combining k-anonymity, l-diversity, and differential privacy to let researchers analyze sensitive patient data without exposing identities, quantifying the privacy-utility trade-off of each technique.

EDUCATION

M.S. in Computer Science - Kent State University, OH

2025

GPA: 3.70 / 4.0